

PAPER_1307 — CKM Matrix Unitarity

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** C — Particle Physics Open (CLOSED) **Date:** June 11, 2026 **Calculator surface:** `calculate_paradox({"paradox": "ckm_unitarity"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1307})`

Master Expression

$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$ via $F_U = 1$ ledger normalization

Match: STRUCTURAL

Interpretation

CKM unitarity emerges from $F_U = 1$ closure. Current measurements show $<10^{-4}$ residual.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $SO_5 = 10$, $A_5 = 60$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$
- $\Lambda_{\text{QCD}} = 0.217 \text{ GeV}$, $v_{\text{Higgs}} = 246 \text{ GeV}$ (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.

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